

RESEARCH ARTICLE

CoPaD-Mark: A Coded Parallelizable Deep Learning-Based Scheme for Robust Image Watermarking

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ABSTRACT In this work, we propose a coded, parallelizable, and deep learning-based image watermarking scheme (CoPaD-Mark). The embedding and extraction layers both undergo iterative training to achieve robust image watermarking. The embedding layer employs a parallel structure with convolutional neural networks inspired by the Inception Net, while the extraction layer uses deformable convolutions. These convolutions enable dynamic feature extraction by adjusting convolutional filters based on image content. The information bits embedded in the image are derived from the parity bits of an error-correcting code. This code's message is constructed by combining a chaotic binary sequence with the original watermark bits. The inclusion of chaotic sequences adds randomness and complexity to the encoding process, enhancing robustness against attempts to alter the watermark. The proposed method produces watermarked images with peak signal-to-noise ratios (PSNR) above 36 dB. CoPaD-Mark is evaluated under several common attacks, including noise addition, filtering, cropping, compression, and resizing, and demonstrates robust performance compared to state-of-the-art methods.

INDEX TERMS Image watermarking, deep learning, chaotic sequences, error correcting codes, convolutional neural networks, robustness.

I. INTRODUCTION

Digital image watermarking refers to the technique of invisibly embedding a message (consisting of watermark bits) within an image (called the original image). Subsequently, the embedded information can be extracted from the watermarked image [1]. This technique serves various purposes, including ownership identification [2], [3], authenticity verification [3], [4], [5], automated monitoring [6], [7], and copyright protection [4], [8], [9]. Watermarking is a vital tool in safeguarding the rights of content creators and copyright holders in the digital domain.

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Digital image watermarking involves two main processes: watermark embedding and watermark extraction [10]. It is categorized into three types: robust, fragile, and semi-fragile [10]. A robust watermarking technique ensures that the watermark bits remain detectable or recoverable even after the watermarked image undergoes various image transformations such as scaling, compression, cropping, filtering [1], [11]. A fragile watermarking is concerned with complete integrity verification. It is designed to detect any modifications to the watermarked image (tamper detection) and to recover the tampered areas (image recovery) [12], [13], [14]. Unlike robust and fragile watermarking, semi-fragile watermarking [15], [16], [17] is designed for tamper detection and image recovery, while still allowing some degree of image transformations. Its main disadvantage is a reduced

recovery rate compared to that achieved by fragile schemes. The extraction process is called blind if it does not require access to the original data, including both watermark bits and the original image.

Additional considerations in watermarking systems include the embedding domain, which can be classified into two main categories: the spatial domain and the frequency domain. In the spatial domain, watermark embedding involves modifying the pixel levels of the original image [12], [15]. In the frequency domain, information can be inserted by manipulating image coefficients using various transformations, including discrete cosine transform (DCT) [8], [18], [19], [20], discrete wavelet transform (DWT) [21], [22], [23], [24], [25], discrete Fourier transform (DFT) [9], [22], [26], [27]. Techniques in the frequency domain are generally more robust but require higher computational complexity compared to spatial domain schemes [8], [28], [29], [30].

Chaotic maps are commonly used to enhance the security of image watermarking schemes [8], [9], [19], [25], [31]. These maps exhibit sensitivity to the initial conditions and pseudo-random behavior, yielding noise-like signals despite being deterministic [12], [17], [32]. To support severe distortions applied to watermarked images, error correction codes can also be employed into the watermarking system [11], [12], [33], [34]. Articles et al. [29] and Ramos et al. [12] have proposed a method where watermark bits are combined with bits generated from a chaotic map using an error-correcting code. The bits embedded into the original image are derived from the parity bits of an error-correcting code. These bits are formed by combining the watermark bits with chaotic bits generated from a secret key shared among all authorized users. The key aspect of the proposed extraction procedure is that it is used solely for retrieving the watermark bits, since the extraction algorithm has knowledge of a portion of the codeword (the chaotic bits), resulting in a highly robust scheme.

Traditional watermarking algorithms often face embedding limitations due to the heuristic nature of the problem. Conversely, deterministic extraction of watermark bits may not be suitable in the presence of attacks. A statistical approach could potentially yield superior performance in such scenarios [30], [35]. As a result, recent research has increasingly explored the use of machine learning tools for watermarking purposes, including K-Means Clustering (KMC) [36], Support Vector Machine (SVM) [37], Bi-directional Extreme Learning Machine (B-ELM) [38], Principal Component Analysis (PCA) [39], Online Sequential Extreme Learning Machine (OSELM) [40], K-Nearest Neighbors (KNN) [41].

Deep learning has emerged as a powerful tool in watermarking due to its capability to learn complex features from data to automatically derive representations based on a minimization of a loss function using training datasets. This enables the development of robust and imperceptible watermarking schemes. Furthermore, deep learning

facilitates the creation of end-to-end watermarking systems, automating the process and potentially improving efficiency compared to conventional techniques. In deep learning watermarking schemes two types of networks commonly employed: Convolutional Neural Networks (CNNs) [30], [35], [42], [43], [44] and Generative Adversarial Networks (GANs) [45], [46].

In this work, we propose a coded, parallelizable, and deep learning-based watermarking scheme, named CoPaD-Mark. Both the embedding and extraction layers undergo iterative training to perform blind and robust watermarking processing on grayscale images operating in the spatial domain. The key features of the proposed scheme include:

- The embedding layer uses a parallel structure with CNNs similar to the Inception Net [47]. This design leverages parallel convolutions to capture complex representations of input data, exhibiting high computational efficiency compared to conventional CNNs blocks [47]. Additionally, the embedding layer incorporates a residual module, which aids faster convergence through improved back-propagation and helps mitigate overfitting issues [48].
- We adopt the coding strategy described in [12], [29] for embedding parity bits of an error-correcting code into the original image. However, our approach differs in that we utilize an unequal error protection (UEP) code. This UEP code prioritizes protection for the first position of a codeword, corresponding to the watermark bits inserted into the image feature space. In the decoder, both the parity bits and chaotic sequences are employed to estimate the watermark bits. By combining this coding approach with UEP, we enhance the resilience of the system, ensuring that errors in the watermark position are detected and corrected with high priority.
- In the extraction layer, we introduce deformable convolutions [49], enabling the convolution window to dynamically adapt and deform based on the image content's characteristics. This adaptation enhances the capture of precise spatial features, thereby improving the extraction of parity bits at the decoder [49].
- Furthermore, we explore the trade-off between imperceptibility and robustness of watermarking by introducing a strength factor. We then conduct performance comparisons of our proposed architecture with existing robust watermarking methods based on deep learning.

The rest of this work is organized into five sections. In Section II, we provide an overview of key concepts related to chaotic maps and error-correcting codes used in the proposed approach. The CoPaD-Mark scheme is detailed in Section III. Section IV discusses the training setup and some metrics used to evaluate the imperceptibility and robustness of watermarking schemes. Experimental results and performance comparisons are presented in Section V. Concluding remarks are summarized in Section VI.

A. LITERATURE REVIEW

In this section, we provide a brief overview of several robust image watermarking schemes based on deep learning proposed in the literature. The rapid growth of artificial intelligence has led to the widespread adoption of deep learning models to address the trade-off between imperceptibility and robustness in robust watermarking systems.

Ahmadi et al. [42] introduce the RedMark network, a deep end-to-end diffusion watermarking framework designed for grayscale images in a transformed space. This framework consists of two fully CNNs structured as auto-encoders with a residual architecture. It incorporates a differentiable attack module within the network to enable end-to-end training. During the extraction process, a voting procedure is employed to estimate the watermark bits. In the robust watermark scheme proposed by Lee et al. [35], CNN serves as the base scheme to preprocess both the original image and the watermark image. It incorporates a CNN-based network for watermark embedding, an attack simulation during training, and a network dedicated to watermark extraction.

Jiang et al. [50] propose the RDD-Net, a robust duplicated-diffusion image watermarking model based on a deep network. This model utilizes an encoder-noise-decoder structure for end-to-end training. Additionally, the scheme incorporates an optimizer mechanism that evaluates similarities between watermarked and original images, thereby enhancing the overall performance of the watermarking scheme. He et al. [51] introduce a feature extraction module integrated with dense connections, enabling the extraction of features from multiple CNN layers to enhance resistance against different image attacks. In Singh and Kumar [30], features from both the original and watermark bits are extracted via an encoder network and subsequently concatenated to produce a watermarked image. Upon reception, a denoising autoencoder network is employed to mitigate noise variations from the watermarked image. Subsequently, watermark bits are extracted using a CNN.

Zhao et al. [43] propose DARI-Mark, a deep learning and attention network tailored for robust image watermarking. Their framework comprises an attention network designed to identify regions that minimally affect the human visual system (HVS). The watermark embedding network incorporates a DCT layer and additional CNN layers. The embedding strength factor controls the intensity of watermark embedding. Jamali et al. [44] present a watermarking scheme that duplicates the watermark string, generating a matrix of the same dimensions as the original image. This strategy facilitates spreading the watermark information across a wider area. Furthermore, they incorporate a dilation convolutional network to introduce redundancy to the watermark bits, thereby enhancing robustness during extraction. The watermark extraction process employs a voting mechanism to recover the watermark.

Alternatively, GANs have also been used for designing deep learning-driven watermarking techniques. In Li et al.'s work [46], a concealed attack method for robust

watermarking based on a generative model and perceptual loss is introduced. In their approach, the generative networks take the watermarked image as input. Inspired by the U-Net architecture, their model consists of an encoder-decoder structure with skip connections. The discriminator's role is to ensure that the distribution of the watermarked image closely matches that of the original image. In the ARWGAN architecture proposed in [45], the discriminator is used to distinguish the watermarked image from the original image. To optimize feature extraction and enhance the learning, a feature fusion module is devised. Additionally, an attention module helps mitigate image distortion caused by the watermark embedding.

In screen-shooting scenarios, watermarking must remain robust against complex distortions, such as grayscale shifts, geometric warping, and sensor noise, introduced when images are captured from a display by external devices like smartphone cameras, which degrade the watermark and challenge robust extraction. To tackle these challenges, Li et al. [52] propose a deep learning-based Screen-Shooting Distortion Simulation (SSDS) framework that models grayscale deviations and camera pipeline noise during training. Fu et al. [53] embeds watermark features in the wavelet domain using a cascade network, and employs a dedicated recovery network to mitigate screen-capture noise.

Recent studies have introduced contrastive learning to enhance the robustness of deep watermarking models. Meng et al. [54] proposed a dual-pathway fusion network for blind visible watermark removal, where contrastive loss encourages the model to disentangle watermark features from background content. Liao et al. [55] designed a screen-shooting robust watermarking method by combining contrastive learning with a mask-guided embedding strategy, effectively improving resilience against display capture distortions. In the medical imaging domain, Liu et al. [56] developed an adversarial contrastive learning-based zero-watermarking approach, which strengthens resistance against adversarial perturbations in volumetric image data.

II. PRELIMINARIES

In this section, we provide a brief introduction to fundamental concepts regarding chaotic maps [57], [58] and error-correcting codes [59], [60]. These concepts are explored further in the robust watermarking algorithm proposed in the next section.

A. CHAOTIC MAPS

The behavior of unidimensional chaotic maps is observed through a discrete time series $\{x_i\}_{i=0}^{\infty}$, which is obtained by iterating a nonlinear and non-invertible function $f(x)$ starting from an initial condition x_0 , as follows [57]

$$x_n = f(x_{n-1}), n = 1, 2, 3, \dots \quad (1)$$

An orbit of x_0 under $f(x)$ is a set of points $\{x_0, f(x_0), f^2(x_0) \dots\}$, where f^k denotes the k -th composition of $f(x)$. Examples of chaotic maps include the cubic map $f(x) =$

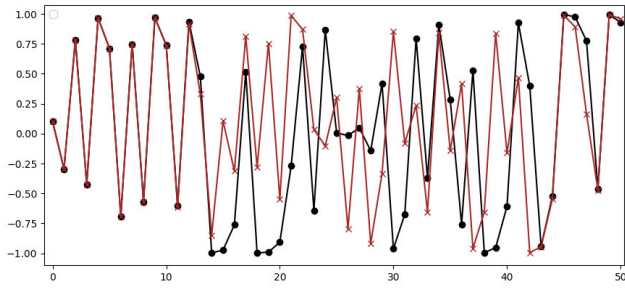


FIGURE 1. Two different orbits for $x_0 = 0.1$ and $x'_0 = 0.100001$. Cubic map.

$4x^3 - 3x$ [58], and the logistic map $f(x) = rx(1 - x)$ [58], where r is a control parameter. Chaotic systems are deeply sensitive to the initial conditions, meaning that infinitesimally close initial conditions generate low correlated orbits. Fig. 1 illustrates this phenomenon with two orbits of the cubic map, generated from initial conditions differing by 10^{-6} . After a few iterations, these orbits display distinct dynamical behavior.

A balanced binary sequence $\{z_n\}$, hereafter referred to as a chaotic binary sequence, is derived from $\{x_n\}$ through a partition of the map domain into two regions, $\mathcal{R}_0$ and $\mathcal{R}_1$, satisfying $\Pr(x_n \in \mathcal{R}_0) = \Pr(x_n \in \mathcal{R}_1) = 1/2$. The sequence $\{z_n\}$ is generated by setting $z_n = 0$ if $x_n \in \mathcal{R}_0$, and $z_n = 1$ if $x_n \in \mathcal{R}_1$.

Chaotic dynamics may also emerge in systems operating in two dimensions. The iteration of the map from an initial condition (x_0, y_0) generates two-dimensional sequences $\{(x_0, y_0), (x_1, y_1), (x_2, y_2) \dots\}$, where $(x_{n+1}, y_{n+1}) = f(x_n, y_n)$. For example, the Arnold cat map (ACM) is defined on the square $f: [0, 1) \times [0, 1) \rightarrow [0, 1) \times [0, 1)$ by

$$f(x, y) = (2x + y, x + y). \quad (2)$$

Its discrete version, the Discrete Arnold Cat Map (DACM), acts on the integer ring modulo Q , $Z_Q \times Z_Q$, and is defined as

$$f(x, y) = (2x + y, x + y) \pmod{Q}. \quad (3)$$

B. ERROR-CORRECTING CODE

Let $C(n, k)$ be a linear binary code, where n is the codeword length and k is the length of the information vector. The generator matrix $\mathbf{G}$, of size $k \times n$, has k linearly independent rows forming a basis for C and is used for encoding. Given an information vector $\mathbf{u} = [u_0, u_1, \dots, u_{k-1}]$, the corresponding codeword $\mathbf{c} = [c_0, c_1, \dots, c_{n-1}]$ is obtained by $\mathbf{c} = \mathbf{u}\mathbf{G}$.

For each generator matrix $\mathbf{G}$, there exists a corresponding parity-check matrix $\mathbf{H}$ of size $(n - k) \times n$. A linear block code is uniquely specified by either $\mathbf{G}$ or $\mathbf{H}$. The rows of $\mathbf{H}$ are linearly independent and orthogonal to those of $\mathbf{G}$. Thus, any valid codeword $\mathbf{c}$ satisfies $\mathbf{c}\mathbf{H}^T = \mathbf{0}$, where $\mathbf{H}^T$ is the transpose of $\mathbf{H}$ and $\mathbf{0}$ represents the zero vector. For example, the matrix $\mathbf{G}$ of the Hamming code $C(15, 11)$ is given by (4),

as shown at the bottom of the next page, and the matrix $\mathbf{H}$ is (5), as shown at the bottom of the next page.

The matrix $\mathbf{G}$ in (4) is in systematic form, $\mathbf{G} = [\mathbf{I}_k | \mathbf{P}]$, where $\mathbf{I}_k$ is a $k \times k$ identity matrix and the $(n - k)$ parity bits are derived from the columns of $\mathbf{P}$. In particular, a codeword in this code is written as $\mathbf{c} = [u_0, u_1, \dots, u_{10}, p_0, p_1, p_2, p_3]$, where p_i are the parity bits. The minimum distance, $d_{\min}$, of a code C refers to the smallest Hamming distance between any pair of distinct codewords. A code can correct errors caused

by patterns with a weight up to or equal to $t = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor$, where $\lfloor \cdot \rfloor$ represents the floor function. This parameter is known as the error correcting capability of the code. For instance, the error correcting capability of the Hamming code is $t = 1$.

UEP codes provide different protection levels to positions in the information vector, extending beyond traditional codes that rely solely on minimum distance. The UEP capability of a code C is measured by the separation vector of length k , denoted as $\mathbf{d} = [d_0, d_1, \dots, d_{k-1}]$ [61], where each element d_j corresponds to the protection level assigned to the j -th component of the information vector and is defined as

$$d_j = \min\{w(\mathbf{u}\mathbf{G}) | u_j \neq 0\}, \quad 0 \leq j \leq k - 1$$

with $w(\cdot)$ denoting the Hamming weight. The smallest element in the separation vector $\mathbf{s}$ is the minimum distance of the code. A code exhibits equal error protection (EEP) capability if all entries of $\mathbf{s}$ are equal; otherwise the code has the UEP property. The Hamming code is an example of an EEP code. Consider a UEP code $C(15, 11)$ with the following matrix $\mathbf{G}$ as presented in [62], (6), as shown at the bottom of the next page.

The matrix $\mathbf{H}$ of this code is (7), as shown at the bottom of the next page. The separation vector is $\mathbf{s} = [5, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2]$. This indicates that up to two errors can occur in any position of the received vector and still allow for correct decoding of the first bit of the information vector. However, positions identified by $d_j = 2$ do not guarantee accurate decoding because they lack sufficient error correction capability.

III. THE COPAD-MARK FRAMEWORK

A new end-to-end robust watermarking architecture (CoPaD-Mark) is proposed in this section. The embedding and extraction networks are iteratively trained to perform blind, robust watermark processing on grayscale images.

The block diagram of the CoPaD-Mark, shown in Fig. 2, comprises three main layers: embedding, attack, and extraction. We adopt the strategy proposed in [12] and [29], where the bits embedding into the $N \times M$ original image C_O are derived from the parity bits p_i of an error-correcting code. The code's information vector is constructed by combining the bits from an $M' \times M'$ binary watermark image W with chaotic bits generated from a secret key. These parity bits are generated by the block Watermark Encoding, but now with a different approach, using a UEP code. The strength

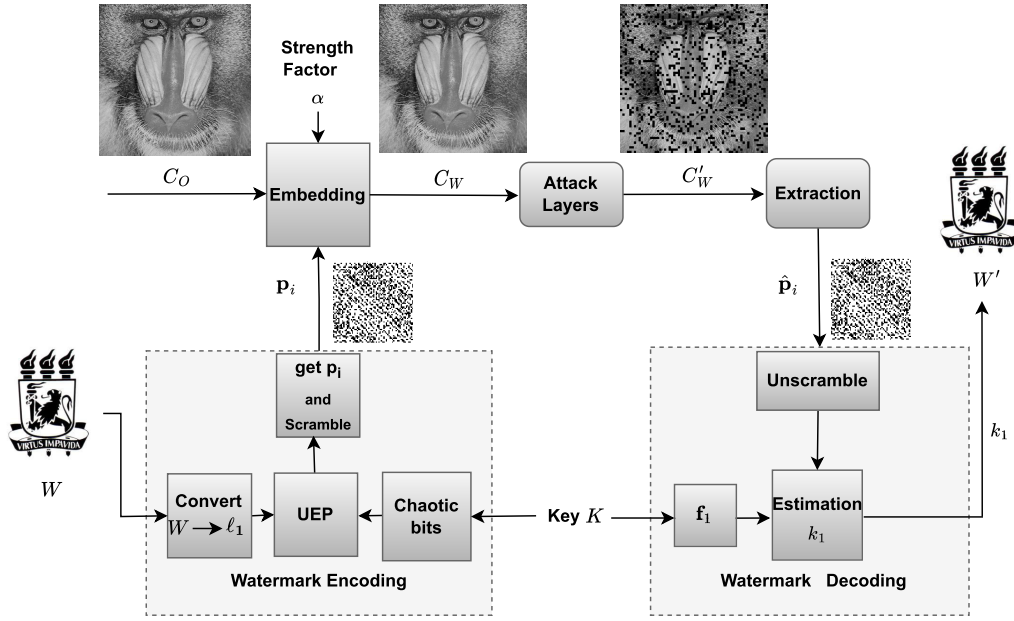


FIGURE 2. Block diagram of the CoPaD-Mark architecture.

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix} \quad (4)$$

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (6)$$

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

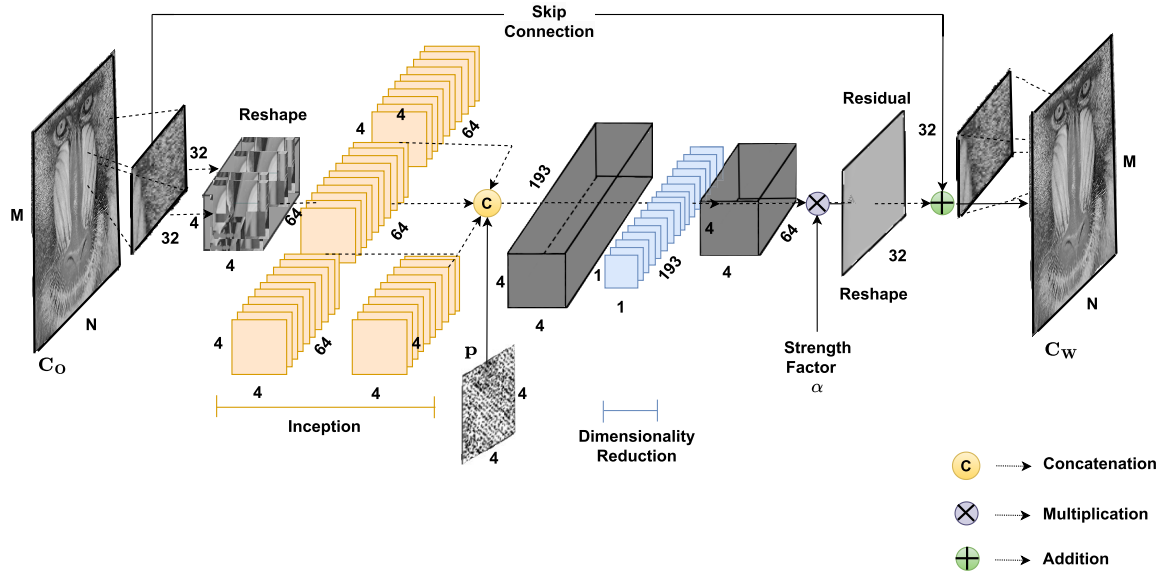


FIGURE 3. Block diagram of the embedding network.

factor α controls the trade-off between imperceptibility and robustness. The watermarked image C_W is subjected to several image attacks, resulting in a corrupted image C'_W , which is the input to the extraction layer. This layer is responsible for extracting the corrupted parity bits $\hat{p}_i$. Finally, a decoding process estimates the watermark bits W' based on a decision rule. Next, we provide the details of the network structure.

A. WATERMARK ENCODING

Watermarking algorithms are required to balance robustness with minimum visual impact. To achieve this goal, we propose to encode the watermark bits. This approach introduces redundancy, enhancing the accuracy of bit pattern extraction. The encoding of the watermark bits is carried out in the following steps.

- 1) Generate a chaotic binary sequence f_1 using the logistic map with the key K .
- 2) Construct the parity check sequence $\mathbf{p}$ of the UEP code $C(15, 11)$ as follows. The binary watermark image of size $M' \times M'$ is converted into a sequence ℓ_1 . The 11 information bits are obtained by concatenating k_1 bits from ℓ_1 and k_2 bits from the chaotic sequence (f_1 sequence), where $k_1 + k_2 = 11$. After encoding, a 15-bit codeword is obtained with 4 parity bits. After repeating this process for the entire ℓ_1 , a parity sequence $\mathbf{p} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_{\lfloor M' \times M' / k_1 \rfloor}\}$ of length $\lfloor 4 \times M' \times M' / k_1 \rfloor$ is obtained. This sequence is then represented as a square image and scrambled with the Arnold cat map. After scrambling, this sequence is embedded into the original image. The use of the UEP code prioritizes protection for the k_1 bits (watermark bits) in the first positions of the codeword.

The steps used to encode the binary watermark using chaotic sequences and UEP coding are summarized in Algorithm 1.

Algorithm 1 Watermark Encoding with UEP and Chaotic Sequences

- 1: **Input:** Binary watermark image W of size $M' \times M'$, secret key K , parameters k_1, k_2 such that $k_1 + k_2 = 11$
- 2: **Output:** Scrambled parity sequence $\mathbf{p}$
- 3: Convert W into a 1D sequence ℓ_1 of length $L = M'^2$
- 4: Generate chaotic binary sequence f_1 of length $\lfloor \frac{L}{k_1} \rfloor k_2$ using logistic map with key K
- 5: Initialize empty parity sequence $\mathbf{p} \leftarrow \emptyset$
- 6: Set total number of codewords $N \leftarrow \lfloor L / k_1 \rfloor$
- 7: **for** $i = 0$ to $N - 1$ **do**
- 8: Extract k_1 bits from $\ell_1[i k_1 : (i + 1) k_1 - 1]$
- 9: Extract k_2 bits from $f_1[i k_2 : (i + 1) k_2 - 1]$
- 10: Concatenate the k_1 and k_2 bits to form an 11-bit message
- 11: Encode the message using UEP code $C(15, 11)$ to obtain a 15-bit codeword
- 12: Extract the 4 parity bits and append to $\mathbf{p}$
- 13: **end for**
- 14: Reshape $\mathbf{p}$ into a square matrix
- 15: Apply Arnold Cat Map scrambling to the parity matrix
- 16: **Return** scrambled parity sequence

B. EMBEDDING LAYER

The main components of the embedding layer comprises the **Inception Net**, **Concatenation**, **Dimension Reduction**, and **Skip Connection** modules, as shown in Fig. 3. This network takes 32×32 patches from the original image as input. Blocks of the parity sequence, each of size 4×4 (obtained from the sequence $\mathbf{p}$) are embedded within each patch.

- **Reshape.** To embed a 4×4 parity block into each 32×32 patch using the proposed architecture, each patch is reshaped into a $4 \times 4 \times 64$ structure.

- **Inception Net.** We employ a parallel structure similar to the Inception Net without a pooling layer, ensuring retention of all information in the image feature maps. The main layer splits into three parallel paths, one path consists of two circular convolutional layers, while the others contain one circular convolutional layer each. Each layer utilizes 64 filters with a kernel size 2×2 , stride equal to 1, and exponential linear unit (ELU) activation functions. The ELU function yields negative values, resulting in outputs close to 0 (on average). This characteristic accelerates the convergence rate during training and mitigates the risk of gradient explosion caused by abrupt changes in derivatives [63]. Circular convolution eliminates the necessity of zero-padding the feature maps, thereby enhancing the strength and robustness of the watermark [42]. These parallel convolution operations allow the network to learn more intricate representations of input data by facilitating both local and global feature extraction. This approach also reduces computational overhead compared to employing a single convolution block with a large number of convolution filters of the same kernel size [47].
- **Concatenation.** The output of the Inception Net (a tensor of the form $4 \times 4 \times ((3 \times 64))$) is concatenated with the 4×4 parity blocks to produce a tensor of dimensions $4 \times 4 \times ((3 \times 64) + 1)$. Injecting the watermark data into the feature space, instead of spreading it across the original image, enhances the network's robustness against various attacks [44].
- **Dimension Reduction.** A dimensional reduction operation is performed using a $1 \times 1 \times 193$ standard convolution layer with 64 filters and the ELU activation function. This operation aims to reduce the dimensions of the output tensor of the concatenation layer to $4 \times 4 \times 64$.
- **Reshape and Skip Connection.** After applying a multiplication operation using the α factor, the resultant tensor is reshaped into a 32×32 form, referred to as the residual. To accelerate the back-propagation process and ensure a comprehensive representation of the patch, a skip connection is implemented. Then, the residual is added to the input patch, resulting in a watermarked patch.
By iterating this process across the image C_O and the sequence $\mathbf{p}$, we obtain the watermarked image C_W .

In the embedding layer, pooling operations are intentionally omitted to preserve fine-grained spatial details crucial for watermark robustness. Instead, dimensionality reduction is performed using a learnable 1×1 convolutional layer, which enables adaptive combination and compression of feature channels while maintaining spatial resolution.

C. ATTACK LAYER

The attack layer takes C_W as input and simulates a variety of attacks. Through the end-to-end training process, this

layer enhances the network's robustness by adjusting its weights and learning to handle various common image transformations. Once trained, the network can effectively manage transformations from classes with similar patterns, including those it is not explicitly trained on. Another benefit of integrating the attack layer during training is its ability to mitigate overfitting and prevent the network from memorizing specific patterns [44]. During each training iteration, one attack is randomly selected from four options: JPEG compression with quality factor $Q = 70$, salt & pepper noise with noise density $s = 0.04$, Gaussian noise with standard deviation $\sigma = 0.3$, and smoothing attack (using a normalized unit mask, though other filters such as a Gaussian or sharpening could be used as well [42]).

D. EXTRACTION LAYER

The extraction module, as depicted in Fig. 4, takes a patch of the watermarked image C'_W as input. Initially, this patch undergoes reshaping. The next module comprises two deformable layers with 64 deformable convolution filters of size 2×2 , along with an ELU activation function to process their feature map. The rationale for employing this convolution stems from the inherent limitation of traditional convolutional networks in modeling geometric transformations due to the fixed geometric structures in their building modules [49]. Deformable convolutions allow the convolutional filters to dynamically adjust their sample positions by incorporating 2D learnable offsets. This capability enhances the network's capacity to capture complex patterns and spatial correlations in the input data. Finally, the output of these deformable convolution blocks is processed by a standard 1×1 convolution layer with one filter and a sigmoid activation function. This layer estimates the block of parity bits $\hat{\mathbf{p}}$.

E. WATERMARK DECODING

The estimated parity sequence $\hat{\mathbf{p}}$ is unscrambled and divided into 4-bit parity sub-sequences, $\hat{\mathbf{p}}_i$. The secret key K is used to replicate the binary chaotic sequence $\mathbf{f}_1$. For each sub-sequences $\hat{\mathbf{p}}_i$, the extraction algorithm knows k_2 chaotic bits, which constitute part of the 11-bit code information message. By fixing these k_2 bits, the remaining k_1 bits determine 2^{k_1} potential parity sequences.

An estimate of the k_1 watermark bits is obtained by computing the smallest Hamming distance between $\hat{\mathbf{p}}_i$ and the possible parity sequences. This process is iterated across the entire $\hat{\mathbf{p}}$ sequence, resulting in the derivation of the estimated watermark image W' .

IV. NETWORK METRICS AND TRAINING

A. LOSSES FUNCTIONS

The structure of the CoPaD-Mark consists of two end-to-end trained networks. The embedding network aims to generate a watermarked image that closely matches the quality of the original image. To achieve this, we initially employ mean square error (MSE) as the loss function to measure the discrepancy between the original and watermarked images,

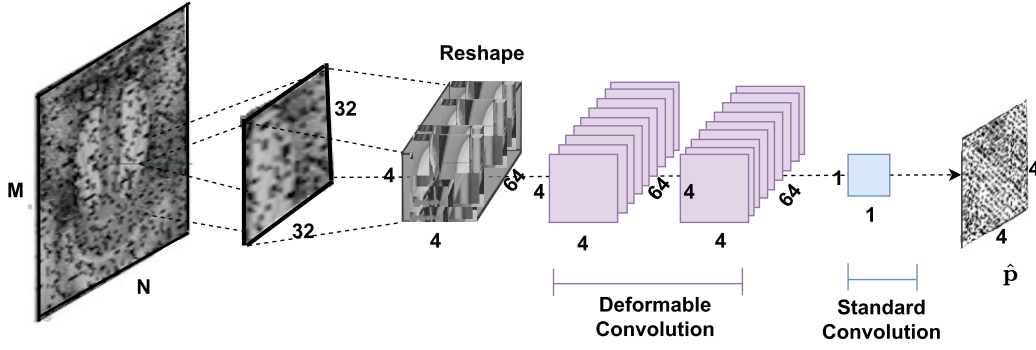


FIGURE 4. Block diagram of the extraction network.

defined as

$$L_1 = \text{MSE} = \frac{1}{M \times N} \sum_{i=1}^N \sum_{j=1}^M (C_O(i, j) - C_W(i, j))^2. \quad (8)$$

The goal of the extraction network is to minimize the disparity between the extracted parity $\hat{\mathbf{p}}$ and the original parity $\mathbf{p}$ used for watermark embedding. To achieve this, we employ the binary cross-entropy L_2 , which is typically applied in binary classification tasks but can be adapted for problems where predicted variables range between zero and one

$$L_2 = - \sum_{\text{all pixels}} y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \quad (9)$$

where y represents the class to be predicted, while $\hat{y}$ denotes the probability assigned to that class. During the end-to-end training, we utilize a weighted combination of L_1 and L_2

$$L_t = \lambda_1 L_1 + \lambda_2 L_2. \quad (10)$$

The hyperparameters λ_1 and λ_2 are determined empirically.

B. IMPERCEPTIBILITY AND ROBUSTNESS METRICS

During the testing phase, several metrics proposed in the literature are employed to evaluate the quality of the watermarked image in terms of imperceptibility and robustness.

The peak signal-to-noise ratio (PSNR) is a measure of watermark imperceptibility, expressed in units of decibels (dB). For 8-bit grayscale images with pixel values from 0 to 255, the PSNR is defined as

$$\text{PSNR} = 10 \log_{10} \left(\frac{255^2}{\text{MSE}} \right) \text{ (dB)} \quad (11)$$

where the MSE is defined in (8). The structural similarity index SSIM is another imperceptibility metric and is defined as

$$\text{SSIM} = \frac{(2\mu_O\mu_W + c_1)(2\rho_{OW} + c_2)}{(\mu_O^2\mu_W^2 + c_1)(\sigma_O^2\sigma_W^2 + c_2)} \quad (12)$$

where μ_O and μ_W are the means of the original and watermarked images, respectively, σ_O^2 and σ_W^2 are the variances of these images, ρ_{OW} is the covariance between C_O and C_W , c_1 and c_2 are fixed constants, $c_1 = 10^{-4}$ and

TABLE 1. Hyperparameters used in training.

Description	Value
Training iteration number	10000
Epochs	63
Strength factor	$\alpha = 1$
Learning rate	10^{-4}
Batch size	32
Adam optimizer	$\beta_1 = 0.900$ $\beta_2 = 0.999$ $\epsilon = 10^{-7}$
Relative loss function weights	$\lambda_1 = 1$ $\lambda_2 = 1$

$c_2 = 9 \times 10^{-4}$. To evaluate the quality of the extracted watermark, we use the bit error rate (BER) defined as

$$\text{BER} = \frac{\sum_{i=1}^{M'} \sum_{j=1}^{M'} (W(i, j) \oplus W'(i, j))}{M' \times N'} \quad (13)$$

where W and W' are the watermark and extracted watermark images, respectively, both of size $M' \times M'$. The symbol $\oplus$ denotes addition modulo 2.

C. TRAINING SETUP

We use the CIFAR-10 dataset [64] and BOSS dataset [65] as original images for network training. The CIFAR-10 dataset comprises 60000 images of size 32×32 pixels, categorized into 10 classes representing various objects. The BOSS dataset consists of 10,000 grayscale images, each sized at 512×512 pixels. We partition these images into patches of size 32×32 pixels, resulting in a collection of 297,011 training patches. These patches are normalized to fall within the range $[-0.5, 0.5]$.

The results are produced using TensorFlow with Keras version 2.10.0, operating with an Anaconda Python environment on a Windows 10 system. The hardware specifications include an Intel(R) Core i9 CPU @ 3.70GHz, 64.0 GB of RAM, and NVIDIA GeForce GTX 3090 GPU. We use the Adam optimizer with a learning rate of 10^{-4} . We also assign equal coefficients for each part of the loss function ($\lambda_1 = \lambda_2 = 1$). Training continues until the total loss function stabilizes at its minimal value. To enhance the algorithm's robustness, we set α to 1.0 during the training phase. Table 1 summarizes the training configu-

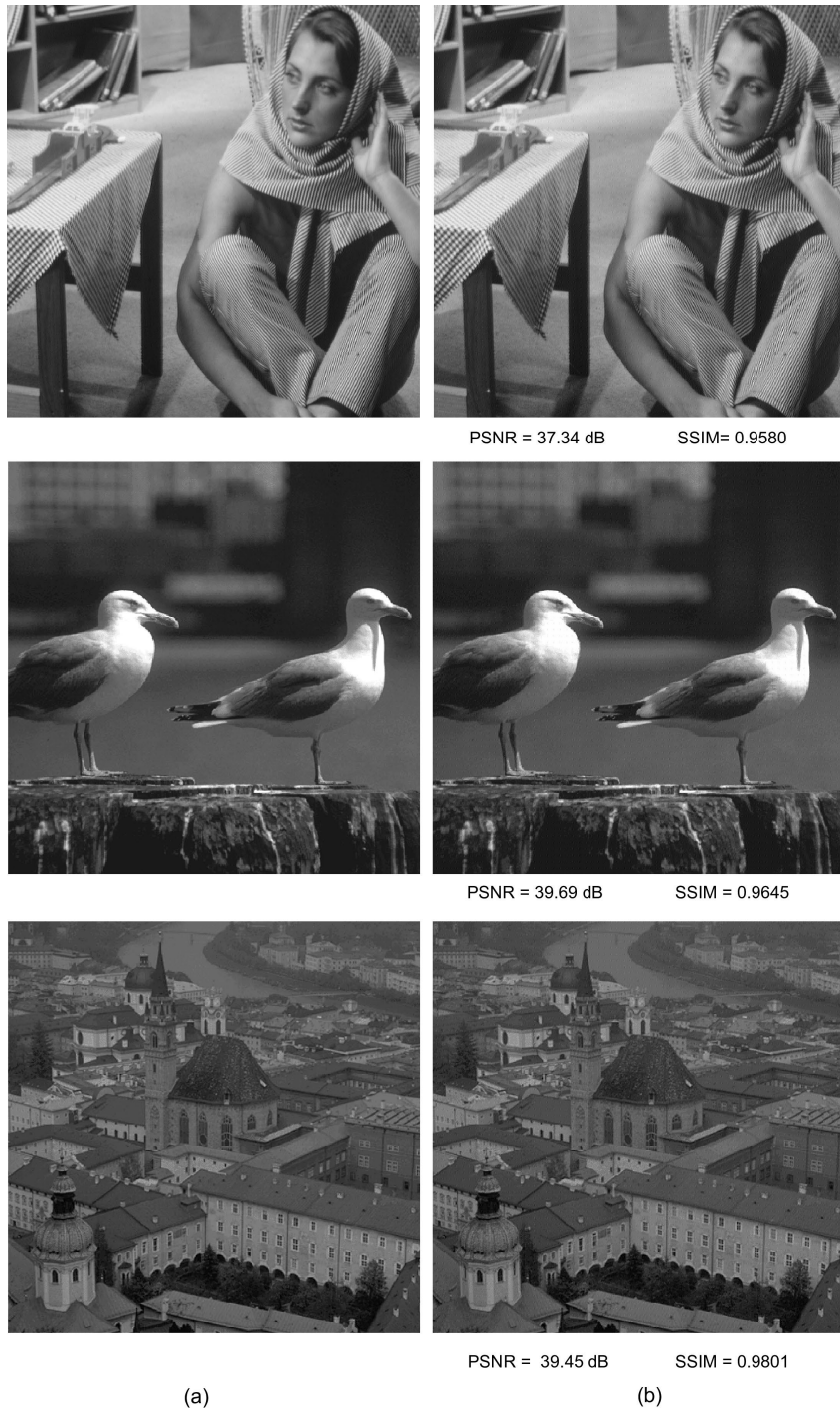


FIGURE 5. Imperceptibility of the watermarked image: (a) C_O original image, (b) C_W watermarked image, with PSNR (dB), and SSIM. $\alpha = 1$.

ration. The approximate training duration is 6 hours and 30 minutes.

V. RESULTS

In this section, we present the results obtained during the testing phase. In this phase, we save the model's weights

and remove the attack layer. We employ the Granada dataset [66], which comprises 49 grayscale images of size 512×512 pixels, as the test set. These images are segmented into 256 non-overlapping patches of size 32×32 pixels. The watermark image W , a random binary image of size 32×32 bits, undergoes an encoding process with parameters

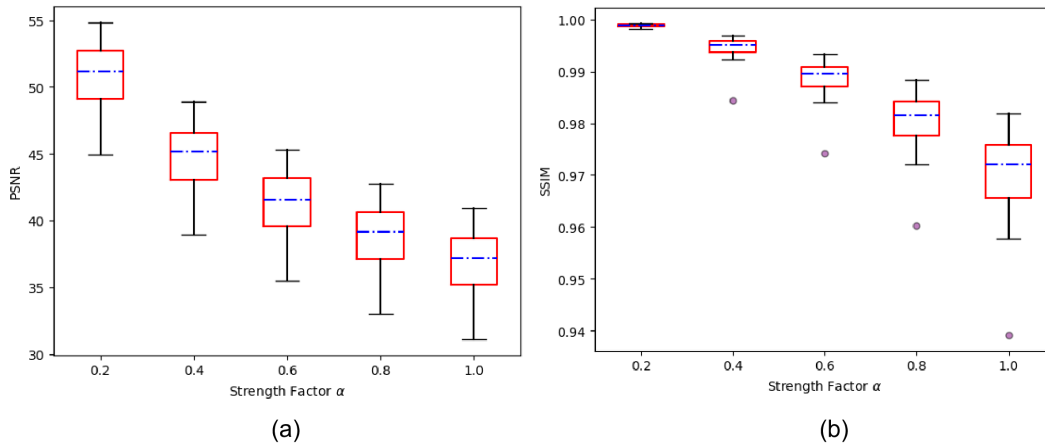


FIGURE 6. Box plot representation of the variation of (a) PSNR (dB) and (b) SSIM, with increasing α .

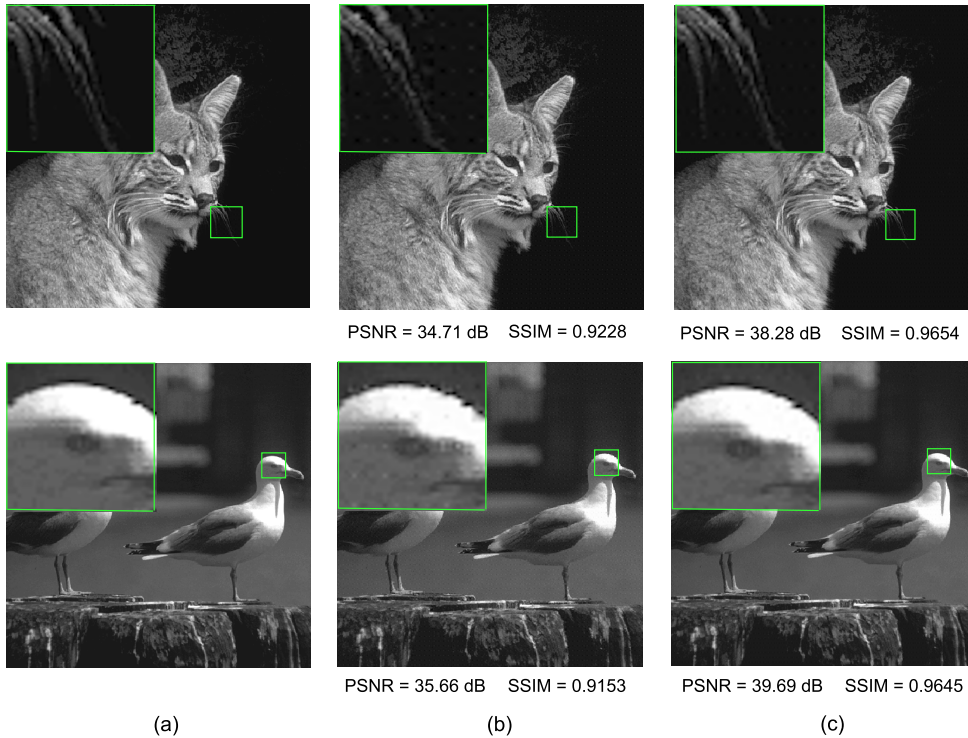


FIGURE 7. Visual difference among: (a) C_O original image, (b) C_W watermarked image (standard convolution), (c) C_W watermarked image (circular convolution) PSNR (dB) and SSIM. $\alpha = 1$.

$k_1 = 1$ and $k_2 = 10$ to yield a parity vector of $32 \times 32 \times 4 = 4096$ bits, segmented into 256 blocks of 4×4 parity bits. Each block is then embedded into one patch of the watermarked image. The generator matrices for the UEP code $C(15,11)$ is given by (6). The watermarked image is subjected to several attacks commonly studied in the literature.

The PSNR and SSIM are measures of image degradation caused by the watermark embedding. The parameter α used in the embedded algorithm modifies the degradation of the original image. Fig. 5 shows the values of PSNR and SSIM for several images, with α equal to 1. The

average PSNR value among all the images in the GRANADA dataset is 36.71dB, with minimum and maximum values in the range of [31.10 dB, 40.94 dB]. The average SSIM is 0.9709, with minimum and maximum values in the range of [0.9578, 0.9818] for the entire dataset. We observe that it is difficult to distinguish between a watermarked image and an original image, indicating effective imperceptibility of the watermarking process.

To assess the influence of the strength factor within the network, we systematically adjust α from 0.2 to 1, observing a decrease in imperceptibility, as indicated by PSNR and

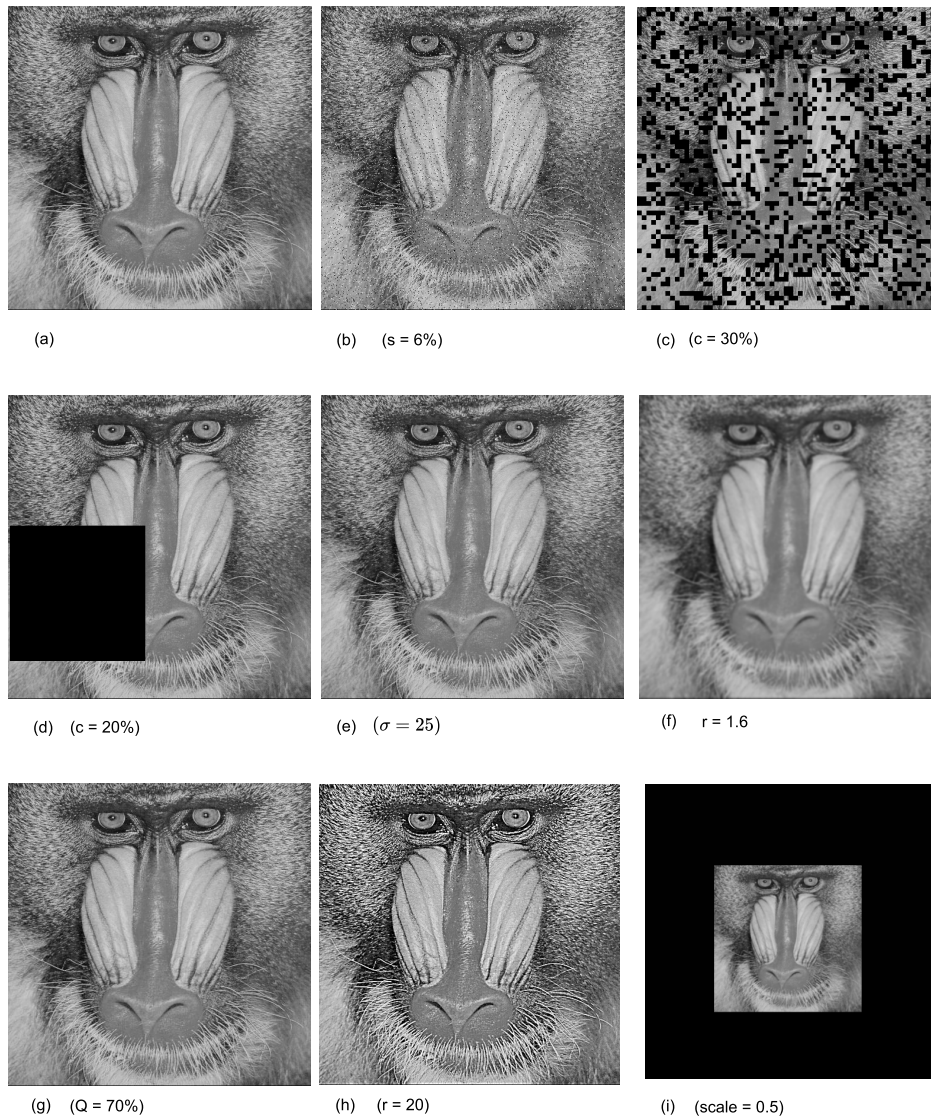


FIGURE 8. Effect of applying a set of attacks on Baboon image: (a) no attack, (b) salt & pepper, (c) grid crop, (d) cropping, (e) Gaussian noise, (f) Gaussian blur, (g) JPEG, (h) sharpening, (i) resizing.

SSIM, across the set of images in the database. The box plots in Fig. 6 visually show the distribution of these metrics for different values of α , providing intuitive insight into the impact of this parameter.

Fig. 7 depicts the discrepancy in visual quality among (a) the original image, (b) the image with standard convolution applied across all embedding layers, and (c) the image with circular convolutions are employed in the Inception Net. Noticeably, the image degradation is less pronounced when employing circular (utilized in CoPaD-Mark) compared to standard convolution. Fig. 8 illustrates a series of attacks applied to the Baboon, including salt & pepper with density s , grid crop and cropping both with percentage c , Gaussian noise with standard deviation σ , Gaussian blur and sharpening with radius r , JPEG compression with quality factor Q , and a resizing attack with parameter scale.

A comprehensive set of imperceptibility and robustness results for CoPaD-Mark is provided Tables 2 and 3, covering various attacks and three values of α . The evaluation encompasses specific images such as Barbara, Bridge, Baboon, and Lena, as well as the Granada dataset. It is observed that increasing the value of α improves the robustness and decreases the imperceptibility. Additionally, for the mean filter attack with a filter size of 5×5 in Table 3, the BER values are around 50 %. Therefore, the proposed system is non-robust in this case.

A. ABLATION EXPERIMENTS

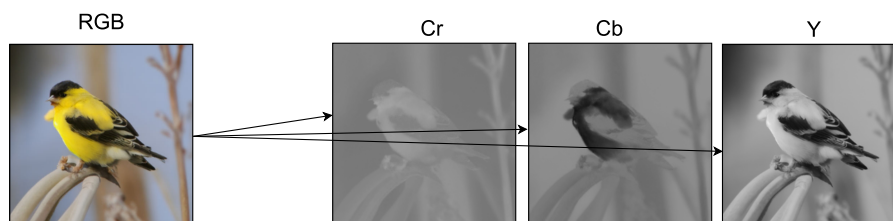
To assess the contribution of key components in the proposed CoPaD-Mark system, we conducted ablation experiments focusing on two aspects: the error protection coding strategy (UEP versus EEP) and the convolutional configuration in the

TABLE 2. Robustness and imperceptibility results for specific images and for the granada dataset.

Figure	α	Imperceptibility		Robustness BER (%)														
		PSNR (dB)	SSIM	Gaussian Noise			Salt & Pepper (%)			Cropping (%)			Grid Crop (%)			JPEG		
				0.05	0.10	0.25	2	6	10	10	20	30	20	30	40	90	70	50
Barbara	1.0	37.34	0.9580	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.46	0.00	2.70	6.10	0.00	2.60	13.81
	0.8	38.72	0.9815	0.09	0.09	0.09	0.09	0.19	0.48	0.58	1.07	2.31	1.52	4.32	6.92	0.97	8.21	22.94
	0.6	39.11	0.9864	0.97	0.97	0.97	1.17	1.56	1.75	1.85	1.75	2.53	3.22	6.73	10.05	4.29	13.27	27.83
Bridge	1.0	35.81	0.9818	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.68	1.17	2.44	6.25	0.00	4.00	10.25
	0.8	37.68	0.9883	0.00	0.00	0.00	0.00	0.09	0.58	0.00	0.87	1.85	1.56	4.78	8.30	0.87	10.64	19.33
	0.6	39.10	0.9915	0.97	0.97	0.97	1.26	1.46	2.44	2.14	4.00	5.85	3.61	6.93	10.93	4.10	17.28	22.07
Baboon	1.0	36.26	0.9799	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.19	0.78	0.48	1.36	6.05	0.00	4.50	12.10
	0.8	37.36	0.9868	0.29	0.29	0.29	0.29	0.39	1.26	0.78	1.17	2.44	1.85	4.00	7.12	0.87	10.05	16.5
	0.6	38.58	0.9901	2.92	2.92	3.02	3.2	2.73	5.17	3.90	5.66	9.27	7.42	7.03	11.71	5.37	17.57	23.53
Lena	1.0	38.64	0.9695	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.68	1.17	3.12	6.15	0.00	0.09	10.25
	0.8	40.58	0.9802	0.00	0.00	0.00	0.00	0.00	0.09	0.19	0.00	2.14	0.68	2.24	6.25	0.09	2.63	20.41
	0.6	41.93	0.9853	0.00	0.00	0.00	0.09	0.09	0.58	0.68	0.97	2.05	1.75	3.71	6.15	0.58	7.32	26.17
Granada	1.0	36.71	0.9709	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.03	1.02	0.84	2.75	5.75	0.00	2.46	11.83
	0.8	38.63	0.9810	0.11	0.11	0.12	0.15	0.23	0.46	0.27	0.48	1.81	1.39	3.49	6.72	0.61	6.67	19.74
	0.6	40.10	0.9860	0.80	0.80	0.80	1.00	1.17	1.87	1.26	1.76	3.35	2.78	5.21	8.44	2.43	12.25	25.08

TABLE 3. Robustness and imperceptibility results for specific images and for the granada dataset.

Figure	α	Imperceptibility		Robustness BER (%)											
		PSNR (dB)	SSIM	Gaussian Filter			Sharpening			Median Filter		Resizing			No Attack
				1	1.6	2.0	1	5	10	3	5	0.5	0.75	1.5	—
Barbara	1.0	37.34	0.9580	0.19	16.50	34.96	0.00	0.00	0.00	1.90	45.3	3.60	0.00	0.00	0.00
	0.8	38.72	0.9815	2.83	20.01	36.91	0.19	0.19	0.39	4.68	46.19	8.10	2.05	1.36	0.09
	0.6	39.11	0.9864	5.76	25.68	43.55	0.97	1.66	2.83	7.21	45.89	10.35	5.07	2.44	0.97
Bridge	1.0	35.81	0.9818	0.29	20.89	37.98	0.00	0.00	0.48	5.85	49.21	5.80	0.50	0.00	0.00
	0.8	37.68	0.9883	4.98	28.22	42.57	0.00	0.39	1.56	9.00	48.04	13.10	3.02	1.85	0.00
	0.6	39.10	0.9915	9.57	31.73	44.33	0.97	3.22	4.10	13.08	48.24	16.99	7.22	3.90	0.97
Baboon	1.0	36.26	0.9799	0.39	20.89	36.81	0.00	0.00	0.00	5.95	49.12	7.10	0.78	0.48	0.00
	0.8	37.36	0.9868	4.29	27.83	37.98	0.48	0.97	0.97	8.59	46.77	12.79	4.39	2.73	0.29
	0.6	38.58	0.9901	8.88	29.78	40.23	2.05	4.58	4.78	14.16	52.24	18.16	10.35	7.91	0.19
Lena	1.0	38.64	0.9695	0.00	11.91	34.08	0.00	0.00	0.00	0.58	46.67	0.97	0.00	0.00	0.00
	0.8	40.58	0.9802	0.39	16.79	33.88	0.00	0.00	0.00	0.97	45.80	1.75	0.19	0.00	0.00
	0.6	41.93	0.9853	1.36	20.21	38.47	0.00	0.19	0.48	1.85	45.99	4.39	0.68	0.00	0.00
Granada	1.0	36.71	0.9709	0.52	16.70	34.50	0.01	0.16	0.37	3.94	52.00	3.76	0.61	0.32	0.00
	0.8	38.63	0.9810	2.82	22.07	38.10	0.10	0.62	1.09	6.38	50.95	7.10	2.19	1.32	0.11
	0.6	40.10	0.9860	5.48	25.91	39.91	0.65	1.73	2.48	9.32	50.68	10.34	4.58	2.84	0.80

**FIGURE 9.** Decomposition of an RGB image into Y (luminance), Cb, and Cr (chrominance) channels in the YCbCr color space.

encoder and decoder. The average BER results, computed over the Granada dataset, are reported in Tables 4 and 5.

First, we evaluated the impact of the convolutional design by comparing CoPaD-Mark configuration, which

TABLE 4. Robustness comparison in terms of the average BER values across the granada dataset. Larger BER values in each comparison are marked in bold.

Granada		circular and deformable $\alpha = 1$	standard convolution $\alpha = 0.68$
Attack	Parameters	PSNR = 36.71 dB SSIM = 0.9709	PSNR = 36.47 dB SSIM = 0.9697
		BER (%)	BER (%)
No Attack	-	0.00	0.42
Salt & Pepper	$s = 6\%$	0.01	1.15
Grid Crop	$c = 30\%$	8.21	3.96
Cropping	$c = 20\%$	0.06	0.87
Gaussian Noise	$\sigma = 25\%$	0.00	0.42
Gaussian Blur	$r = 1.6$	20.14	22.39
JPEG	$Q = 70\%$	2.87	4.11
Sharpening	$r = 20$	0.93	1.85
Resizing	scale = 0.5	4.97	10.86

uses circular convolutions in the embedding layer and deformable convolutions in the extraction layer, with a baseline using standard convolutions in all layers. To ensure a fair comparison, the value of α was adjusted in each setting to match average PSNR and SSIM levels. As shown in Table 4, the CoPaD-Mark configuration yields consistently lower BER, confirming the effectiveness of the proposed convolutions modules in enhancing robustness.

Second, we analyzed the effect of using UEP versus EEP coding. UEP improves robustness by assigning higher redundancy to positions corresponding to watermark bits. The generator matrices for these codes are described in Section II-B. As shown in Table 5, UEP consistently achieves lower BER across most attack types. For instance, under the grid crop distortion, UEP reduces the BER from 8.21 % to 2.75%.

B. COMPARISON WITH STATE-OF-THE-ART

The performance of CoPaD-Mark is compared with the state-of-the-art methods that use end-to-end deep learning technique [35], [42], [43], [44]. We adjust the parameter α to evaluate their robustness with a similar average PSNR for the Granada dataset, ensuring a fair comparison. Table 6 presents a comparison with [42], [43] using $\alpha = 1$ in all cases. The method outlined in [44] does not consider a strength factor in its architecture, yielding an average PSNR of 40.34 dB for the Granada dataset, while in [35], the average PSNR reaches 40.48 dB for $\alpha = 1$. A comparison with these methods is presented in Table 7, where CoPaD-Mark is evaluated with $\alpha = 0.6$, yielding an average PSNR of 40.10 dB. From these tables, we observe that CoPaD-Mark outperforms other methods in some attack scenarios, such as Gaussian noise, salt & pepper, cropping, grid crop, Gaussian filter, resizing. It is important to highlight that while the proposed system is trained for 4 specific attacks, it is also resilient against various

TABLE 5. Robustness comparison between EEP and UEP codes in terms of the average BER values across the granada dataset. Larger BER values in each comparison are marked in bold. $\alpha = 1$.

Granada		EEP	UEP
Attack	Parameters	PSNR = 36.71 dB SSIM = 0.9709	PSNR = 36.47 dB SSIM = 0.9697
		BER (%)	BER (%)
No Attack	-	0.00	0.00
Salt & Pepper	$s = 6\%$	0.01	0.00
Grid Crop	$c = 30\%$	8.21	2.75
Cropping	$c = 20\%$	0.06	0.03
Gaussian Noise	$\sigma = 25\%$	0.00	0.00
Gaussian Blur	$r = 1.6$	20.14	16.70
JPEG	$Q = 70\%$	2.87	2.46
Sharpening	$r = 20$	0.93	0.74
Resizing	scale = 0.5	4.97	3.76

other types of attacks. Overall, CoPaD-Mark is competitive with existing literature in terms of robustness.

To further validate the effectiveness of CoPaD-Mark, we compare its robustness against a classical, non-deep learning watermarking method that leverages transform-domain techniques. Specifically, we consider the approach from [11], which utilizes Reed–Solomon codes combined with DCT. Table 8 summarizes the BER of CoPaD-Mark and this classical method for specific images under various attacks. These comparisons, conducted on images of size 512×512 with a binary watermark of size 32×32 , indicate that CoPaD-Mark offers competitive or improved robustness in several scenarios.

C. TIME COMPLEXITY

We now present the testing time of our system for real-time processing of 512×512 grayscale images from the Granada dataset. When utilizing the GPU, the average insertion time is 0.0032 seconds, and the average extraction time is 0.0024 seconds. Without the GPU, there is an increase of 0.0558 seconds in the insertion time and 0.0266 seconds in the extraction time. Table 9 summarizes the execution times of the CoPaD-Mark architecture for our hardware specifications and compares them with those obtained using other watermarking methods [42], [44]. Comparing the execution times, CoPaD-Mark is faster, but this is achieved using more advanced hardware specifications. Nonetheless, its time complexity remains competitive.

D. ANALYSIS FOR COLOR IMAGES

We evaluate the performance of the CoPaD-Mark architecture for color images in RGB format, focusing on imperceptibility and robustness. To this end, we convert each RGB image into its YCbCr representation, which separates luminance (Y)

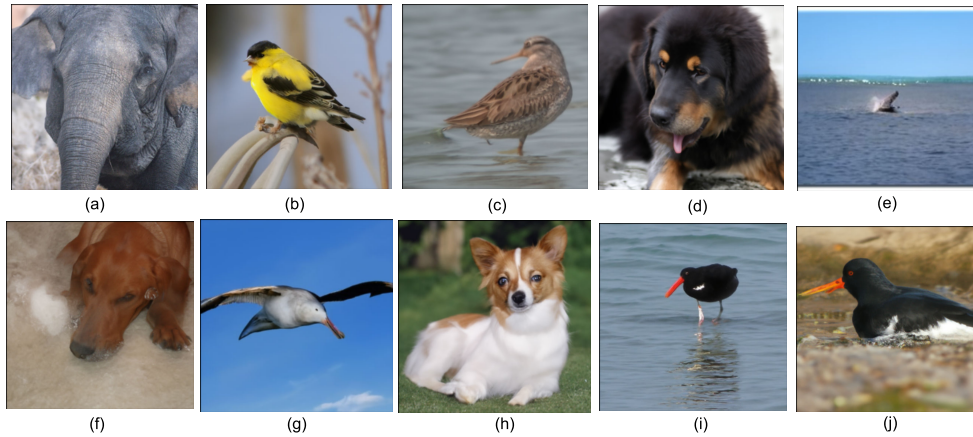


FIGURE 10. Examples of images from the ADM dataset of GenImage.

TABLE 6. Average BER comparison across the granada dataset for $\alpha = 1$. The best values are indicated in bold.

Attack		CoPaD-Mark PSNR = 36.71 dB BER (%)	[42] PSNR = 35.93 dB BER (%)	[43] PSNR = 37.38 dB BER (%)
Gaussian Noise	$\sigma = 0.05$	0.00	0.00	0.09
	$\sigma = 0.15$	0.00	2.60	3.71
	$\sigma = 0.25$	0.00	12.70	10.30
Salt & Pepper	$s = 0.02$	0.00	0.00	-
	$s = 0.04$	0.00	-	0.00
	$s = 0.06$	0.00	0.10	0.00
	$s = 0.10$	0.04	0.90	0.00
Cropping	$c = 10\%$	0.00	6.00	0.68
	$c = 20\%$	0.03	11.30	3.81
	$c = 30\%$	1.02	17.10	4.05
Grid Crop	$c = 20\%$	0.84	3.30	-
	$c = 30\%$	2.75	6.90	-
	$c = 40\%$	5.75	11.40	-
JPEG	$Q = 90$	0.00	0.00	0.00
	$Q = 70$	2.43	0.00	0.00
	$Q = 50$	11.83	1.20	1.36
Gaussian Filter	$r = 1.0$	0.52	0.10	-
	$r = 1.6$	16.70	29.00	-
Sharpening	$r = 1$	0.01	0.00	0.00
	$r = 5$	0.16	0.10	0.00
	$r = 10$	0.37	0.20	0.00
	$r = 20$	0.74	-	0.19
Median Filter	3×3	3.94	2.90	-
Resizing	scale = 0.50	3.76	5.90	14.4
	scale = 0.75	0.61	0.00	5.56
	scale = 1.50	0.32	0.00	0.48
No Attack	—	0.00	0.00	-

from chrominance components (Cb and Cr), as illustrated in Fig. 9. The training phase follows the procedure outlined in Subsection IV. We embed the watermark exclusively into the Y channel during the testing phase. This strategy is supported by previous studies [35], [68], [69], [70], which highlight the benefits of Y-channel embedding in preserving perceptual quality.

We conducted a direct comparison with the scheme proposed in [67], which is specifically designed for

deep learning-based watermarking on color images. This method introduces a bidirection-interactive and context-aware (BICA) network architecture, where encoder and decoder modules exchange features to improve watermark embedding and extraction performance. The watermark bits are randomly generated following a uniform distribution and encoded using BCH codes to improve robustness. We adopted the same test configuration described in [67], selecting 10 random images from the ADM subset of the

TABLE 7. Average BER comparison across the granada dataset. The best values are indicated in bold.

Attack		CoPaD-Mark $\alpha = 0.6$ PSNR = 40.10 dB BER (%)	[35] $\alpha = 1$ PSNR = 40.58 dB BER (%)	[44] PSNR = 40.34 dB BER (%)
Gaussian Noise	$\sigma = 0.05$	0.80	5.99	5.10
	$\sigma = 0.15$	0.80	27.00	32.70
	$\sigma = 0.25$	0.80	38.17	40.50
Salt & Pepper	$s = 0.02$	1.00	1.02	2.80
	$s = 0.04$	1.03	1.53	-
	$s = 0.06$	1.17	3.19	18.50
	$s = 0.10$	1.87	-	34.10
Cropping	$c = 10$	1.26	2.14	2.50
	$c = 20$	1.76	5.33	6.60
	$c = 30$	3.35	8.67	11.10
Grid Crop	$c = 20$	2.78	-	0.10
	$c = 30$	5.21	-	0.40
	$c = 40$	8.44	-	0.50
JPEG	$Q = 90$	2.43	0.96	6.20
	$Q = 70$	12.25	4.24	29.30
	$Q = 50$	25.08	8.06	37.90
Gaussian Filter	$r = 1.0$	5.48	7.14	28.10
	$r = 1.6$	25.91	9.73	41.60
	$r = 2.0$	39.91	12.72	47.30
Sharpening	$r = 1$	0.65	0.99	0.10
	$r = 5$	1.73	1.72	0.80
	$r = 10$	2.48	2.00	1.40
Median Filter	3×3	9.32	8.42	37.30
	5×5	50.68	10.55	49.10
Resizing	scale = 0.50	10.34	-	39.00
	scale = 0.75	4.58	-	25.00
	scale = 1.50	2.84	-	4.80
No Attack		0.80	0.70	-

TABLE 8. Comparison of imperceptibility and robustness under various attacks for two test images. The best values are indicated in bold.

Image		Barbara		Peppers	
Method		CoPaD-Mark $\alpha = 0.55$	[11]	CoPaD-Mark $\alpha = 0.55$	[11]
Imperceptibility	PSNR (dB)	42.27	42.26	45.03	43.36
	SSIM	0.9923	0.9798	0.9901	0.9823
Attack		BER (%)		BER (%)	
Gaussian Noise	$\sigma = 0.001$	1.46	1.85	1.85	2.73
Salt & Pepper	$s = 0.001$	1.26	1.46	1.07	1.39
Cropping(%)	$c = 10$	4.69	1.17	4.20	1.56
	$c = 15$	5.76	6.02	4.39	7.33
	$c = 25$	6.45	11.15	5.47	14.05
Histogram		2.54	7.58	0.39	6.17
JPEG	$Q = 80$	4.88	0.00	3.90	0.00
	$Q = 70$	12.69	0.13	10.74	0.18
	$Q = 60$	13.47	0.77	15.62	0.24
Gaussian Filter	3×3	19.62	0.19	19.53	0.69
Sharpening	$r = 1$	0.68	0.11	0.10	0.33
Median Filter	3×3	12.20	12.35	13.57	13.71
Resizing	scale = 0.80	6.83	7.58	19.53	30.81

GenImage dataset [71], each resized to a resolution of 512×512 pixels. Sample images from this dataset are shown in Fig. 10. The average PSNR and BER results reported in

Table 10 indicate that CoPaD-Mark achieves competitive performance on color images, highlighting its robustness and generalization capabilities.

TABLE 9. Comparison of execution times.

Method	Time (seg)	CPU	GPU	RAM	Software
CoPaD-Mark	0.0056 (0.0032 + 0.0024)	Intel Core i9 3.70 GHz	Yes	64GB	Spyder (Python)
CoPaD-Mark	0.088 (0.059 + 0.029)	Intel Core i9 3.70 GHz	No	64GB	Spyder (Python)
[42]	0.0079 (0.0048 + 0.0031)	Intel Core i7 3.50 GHz	Yes	64GB	Spyder (Python)
[44]	0.0083 (0.0051 + 0.0032)	Intel Core i7 3.50 GHz	Yes	64GB	Spyder (Python)

TABLE 10. Comparison of imperceptibility and robustness under various attacks for 10 random color images. The best values are indicated in bold.

Method		CoPaD-Mark $\alpha = 1$	[67]
Imperceptibility	PSNR (dB)	41.88	41.15
Attack		BER (%)	
Gaussian Noise	$\sigma = 0.1$	0.00	5.17
Salt & Pepper	$s = 0.05$	1.70	3.25
	$s = 0.10$	2.90	14.70
	$s = 0.15$	7.52	17.71
Cropping	$c = 0.3$	1.60	0.01
Grid Crop	$c = 0.3$	3.07	0.01
Gaussian Blur	$r = 2.0$	5.17	1.09
Brightness	$r = 1.1$	0.00	0.00
	$r = 1.2$	0.00	1.59
	$r = 1.3$	0.00	1.89
Contrast	$r = 1.0$	0.00	0.00
	$r = 1.5$	1.22	1.52
	$r = 2.0$	1.78	1.84
Saturation	$d = 15$	0.00	0.00
	$d = 20$	0.00	0.01
	$d = 25$	0.00	1.19
JPEG	$Q = 50$	18.84	0.83
Median Filter	3×3	2.94	0.00
	5×5	28.34	0.00
No Attack	————	0.00	0.00

VI. CONCLUSION

The CoPaD-Mark scheme successfully integrates a parallelizable deep learning approach with coding techniques to create a robust image watermarking scheme. By employing iterative training, parallel structures with CNNs, and deformable convolutions, the system achieves dynamic and resilient feature extraction. The innovative use of UEP codes in conjunction with chaotic sequences ensures robustness against errors, prioritizing the protection of critical watermark bits. Performance comparisons are conducted by adjusting the parameter α to ensure a fair comparison based on similar average PSNR values for the Granada dataset. The results indicate that CoPaD-Mark is effective and competitive with current state-of-the-art methods, confirming its potential as a powerful tool for robust image watermarking. To broaden the practical applicability of CoPaD-Mark, we have extended the evaluation to color images, indicating that the method generalizes well beyond grayscale data. Future work includes developing deep learning schemes specifically tailored for fragile watermarking. Additionally, investigating lightweight optimization techniques such as quantization and pruning

can enhance CoPaD-Mark's applicability in real-time and resource-constrained scenarios. Another important direction is replacing the current fixed patch extraction strategy with dynamic or content-adaptive methods to further improve robustness against geometric distortions.

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